

Like most systems, a living cell is not in equilibrium. Materials flow in and out, keeping metabolic pathways from ever reaching equilibrium, and the cell continues to do work throughout its life. This principle is illustrated by the open (and more realistic) hydroelectric system in **Figure 8.8a**. However, unlike this simple single-step system, a catabolic pathway in a cell releases free energy in a series of reactions. An example is cellular respiration, illustrated by analogy in **Figure 8.8b**. Some of the reversible reactions of respiration are constantly “pulled” in one direction—that is, they are kept out of equilibrium. The key to maintaining this lack of equilibrium is that the product of a reaction does not accumulate but instead becomes a reactant in the next step; finally, waste products are expelled from the cell. The overall sequence of reactions is kept going by the huge free-energy difference between glucose and O_2 at the top of the energy “hill” and CO_2 and H_2O at the “downhill” end. As long as our cells have a steady supply of glucose or other fuels and oxygen and are able to expel waste products to the surroundings, their metabolic pathways never reach equilibrium and can continue to do the work of life.

Organisms are open systems. Sunlight provides a daily source of free energy for an ecosystem’s plants and other photosynthetic organisms. Animals and other nonphotosynthetic organisms in an ecosystem must have a source of free

energy—the organic products of photosynthesis. Now we are ready to see how a cell actually performs the work of life.

CONCEPT CHECK 8.2

1. Cellular respiration uses glucose and O_2 , which have high levels of free energy, and releases CO_2 and H_2O , which have low levels of free energy. Is cellular respiration spontaneous or not? Is it exergonic or endergonic? What happens to the energy released from glucose?
2. **VISUAL SKILLS** How would the processes of catabolism and anabolism relate to Figure 8.5c?
3. **WHAT IF?** Some partygoers wear glow-in-the-dark necklaces that start glowing once they are “activated” by snapping the necklace. This allows two chemicals to react and emit light in the form of chemiluminescence. Is the reaction exergonic or endergonic? Explain.

For suggested answers, see Appendix A.

CONCEPT 8.3

ATP powers cellular work by coupling exergonic reactions to endergonic reactions

A cell does three main kinds of work:

- **Chemical work**, the pushing of endergonic reactions that would not occur spontaneously, such as the synthesis of polymers from monomers (chemical work will be discussed further here; examples are shown in Chapters 9 and 10)
- **Transport work**, the pumping of substances across membranes against the direction of spontaneous movement (see Concept 7.4)
- **Mechanical work**, such as the beating of cilia (see Concept 6.6), the contraction of muscle cells, and the movement of chromosomes during cellular reproduction

A key feature in the way cells manage their energy resources to do this work is **energy coupling**, the use of an exergonic process to drive an endergonic one. ATP is responsible for mediating most energy coupling in cells, and in most cases it acts as the immediate source of energy that powers cellular work.

➔ **Mastering Biology Animation: Energy Coupling**

The Structure and Hydrolysis of ATP

ATP (adenosine triphosphate; see Concept 4.3) contains the sugar ribose, with the nitrogenous base adenine and a chain of three phosphate groups (the triphosphate group) bonded to it (**Figure 8.9a**). In addition to its role in energy coupling, ATP is also one of the nucleoside triphosphates used to make RNA (see Figure 5.23).

▼ **Figure 8.8** Equilibrium and work in open systems.

